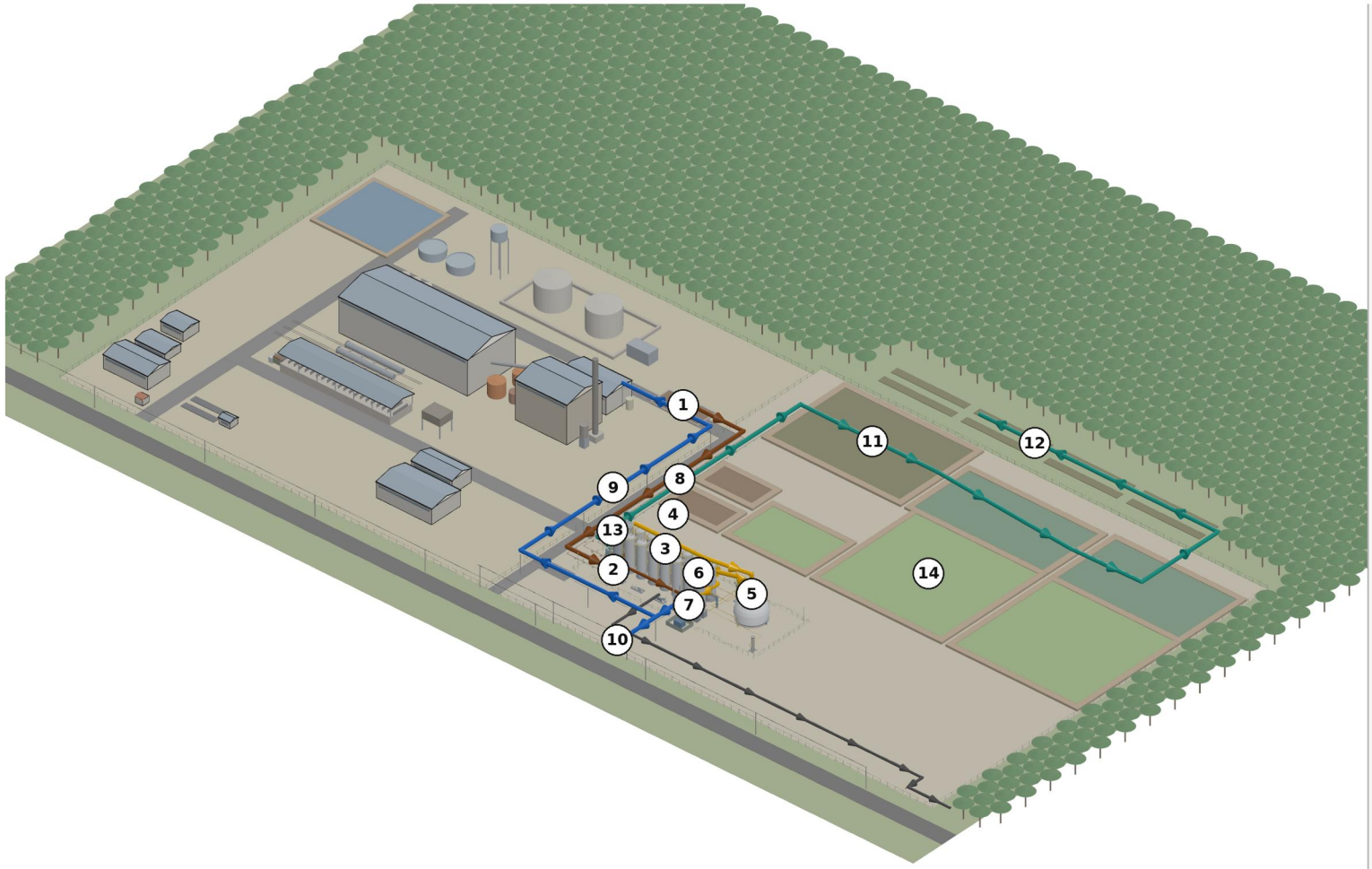
Lokasi PKS: alir POME -> energi



ALIR: POME -> ENERGI PKS

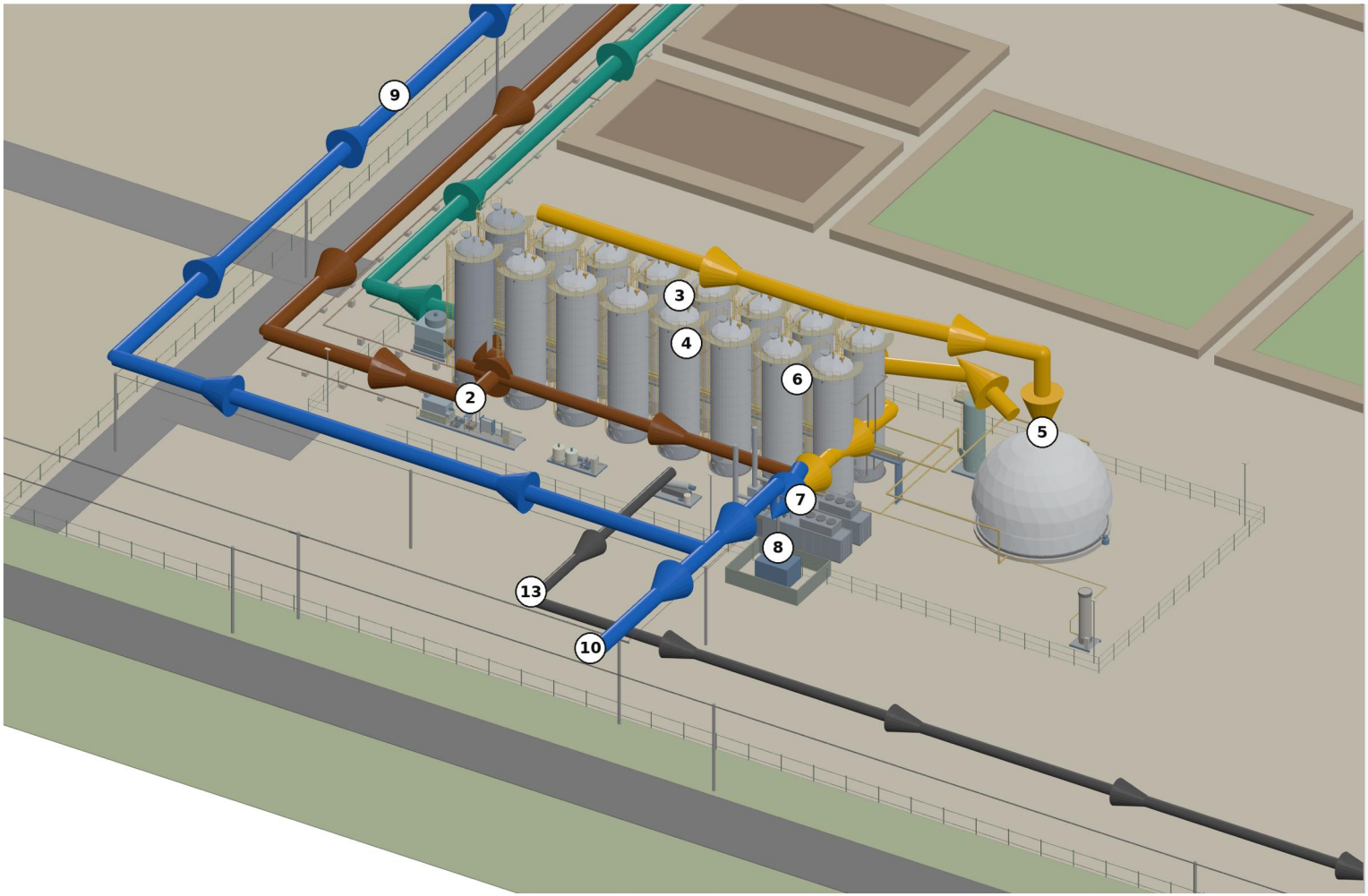
PKS proksi 45 t TBS/jam + set generator biogas portabel

- **Air limbah POME**
- **Biogas**
- **Listrik**
- **Efluen terolah**
- **Kue lumpur**

1. Fat pit PKS: POME panas 85 °C, 585 m³/hari
2. PM-1 praolah: saringan, desander, pendingin 85 -> 56 °C
3. 14 reaktor AnMBR termofilik 55 °C, HRT 4,6 hari
4. PM-4 membran UF 996 m²: biomassa kembali ke reaktor
5. GH-301 penampung biogas 1.135 m³ (11.815 Nm³/hari, CH4 58,5%)
6. PM-6 pembersih biogas: H2S 2.887 -> 34 ppm
7. PM-8 genset 2 x 800 kWe: 1.088 kWe kotor
8. Gardu 0,4/20 kV: 961 kWe neto
9. Listrik ke rumah mesin PKS (0,37 GWh/th)
10. Ekspor ke jaringan PLN 20 kV (5,69 GWh/th)
11. Efluen 541 m³/hari ke kolam polishing
12. Aplikasi lahan ke kebun sawit (flatbed)
13. Kue lumpur 5.708 t/th ke kebun (bahan organik)
14. Kolam anaerobik lama dibebaskan 1,1 ha; emisi neto 32.288 t CO2e/th dihindari

Garis alir skematis (diperbesar), bukan diameter pipa nyata.

Alir di dalam set portabel



ALIR DI DALAM SET PORTABEL

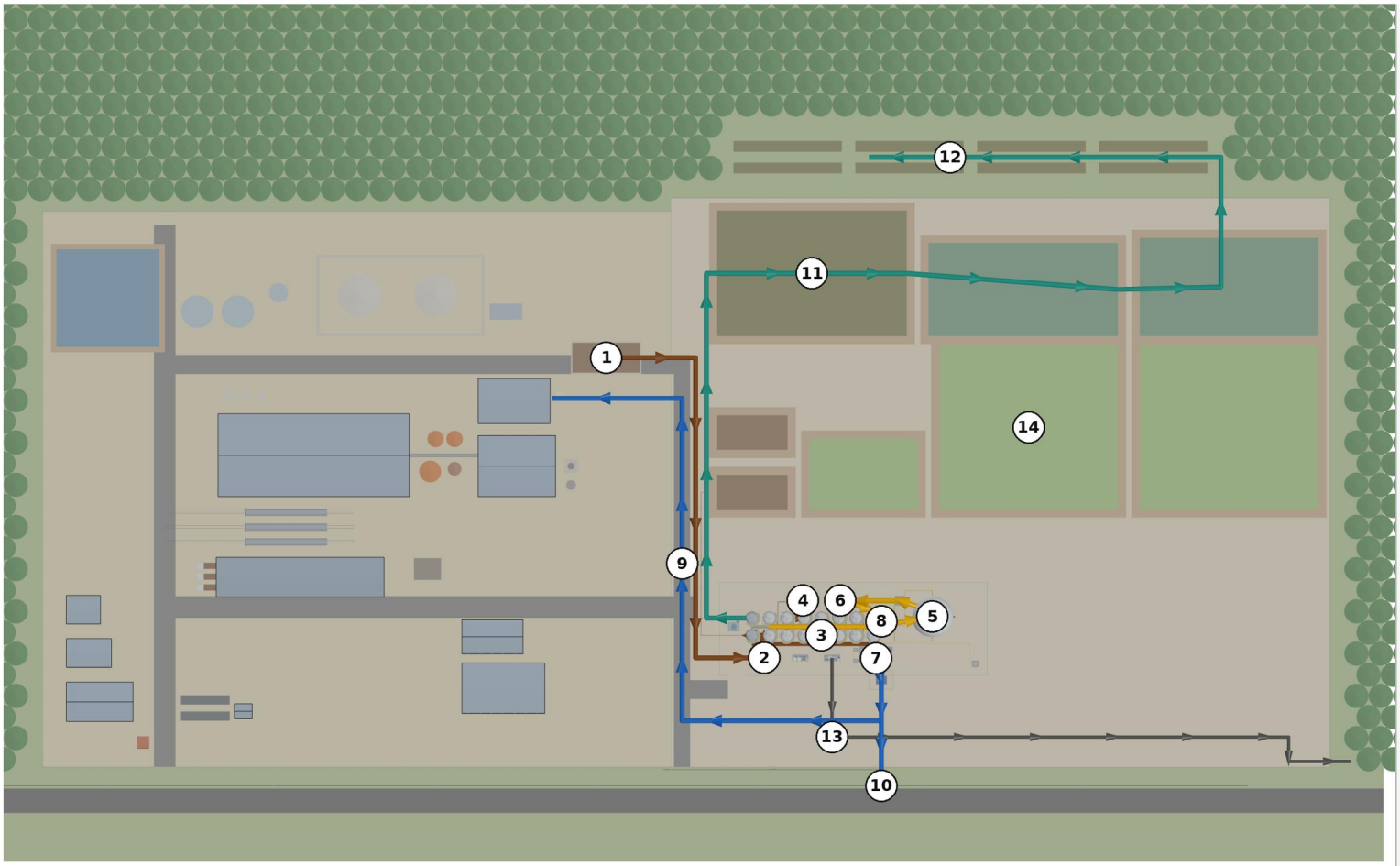
Dari POME panas ke listrik untuk PKS dan PLN

- **Air limbah POME**
- **Biogas**
- **Listrik**
- **Efluen terolah**
- **Kue lumpur**

1. Fat pit PKS: POME panas 85 °C, 585 m³/hari
2. PM-1 praolah: saringan, desander, pendingin 85 -> 56 °C
3. 14 reaktor AnMBR termofilik 55 °C, HRT 4,6 hari
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11. Efluen 541 m³/hari ke kolam polishing
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13. Kue lumpur 5.708 t/th ke kebun (bahan organik)
14. Kolam anaerobik lama dibebaskan 1,1 ha; emisi neto 32.288 t CO2e/th dihindari

Garis alir skematis (diperbesar), bukan diameter pipa nyata.

Denah alir lokasi



DENAH ALIR LOKASI

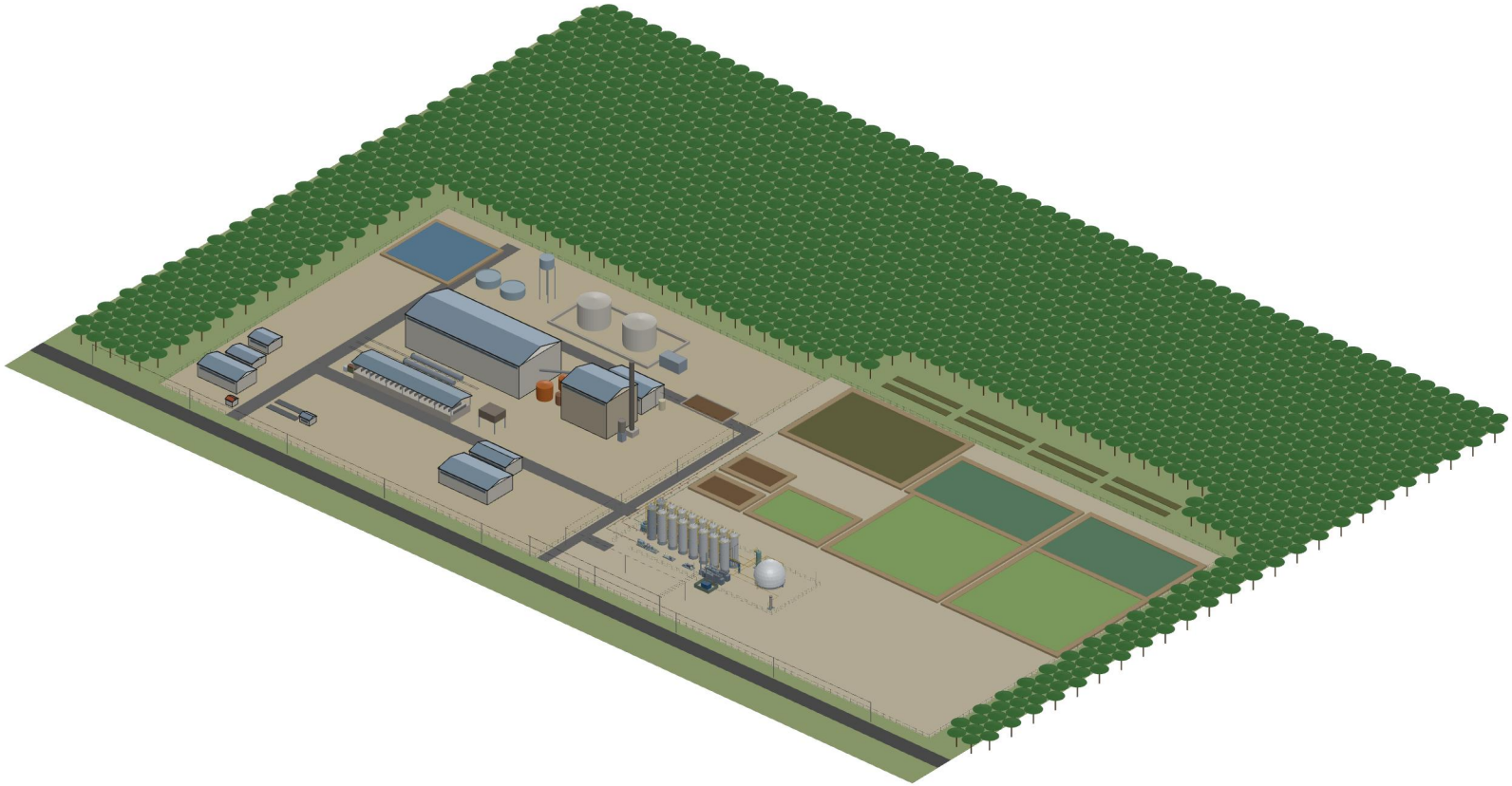
Tampak atas, skala tidak tercantum

- **Air limbah POME**
- **Biogas**
- **Listrik**
- **Efluen terolah**
- **Kue lumpur**

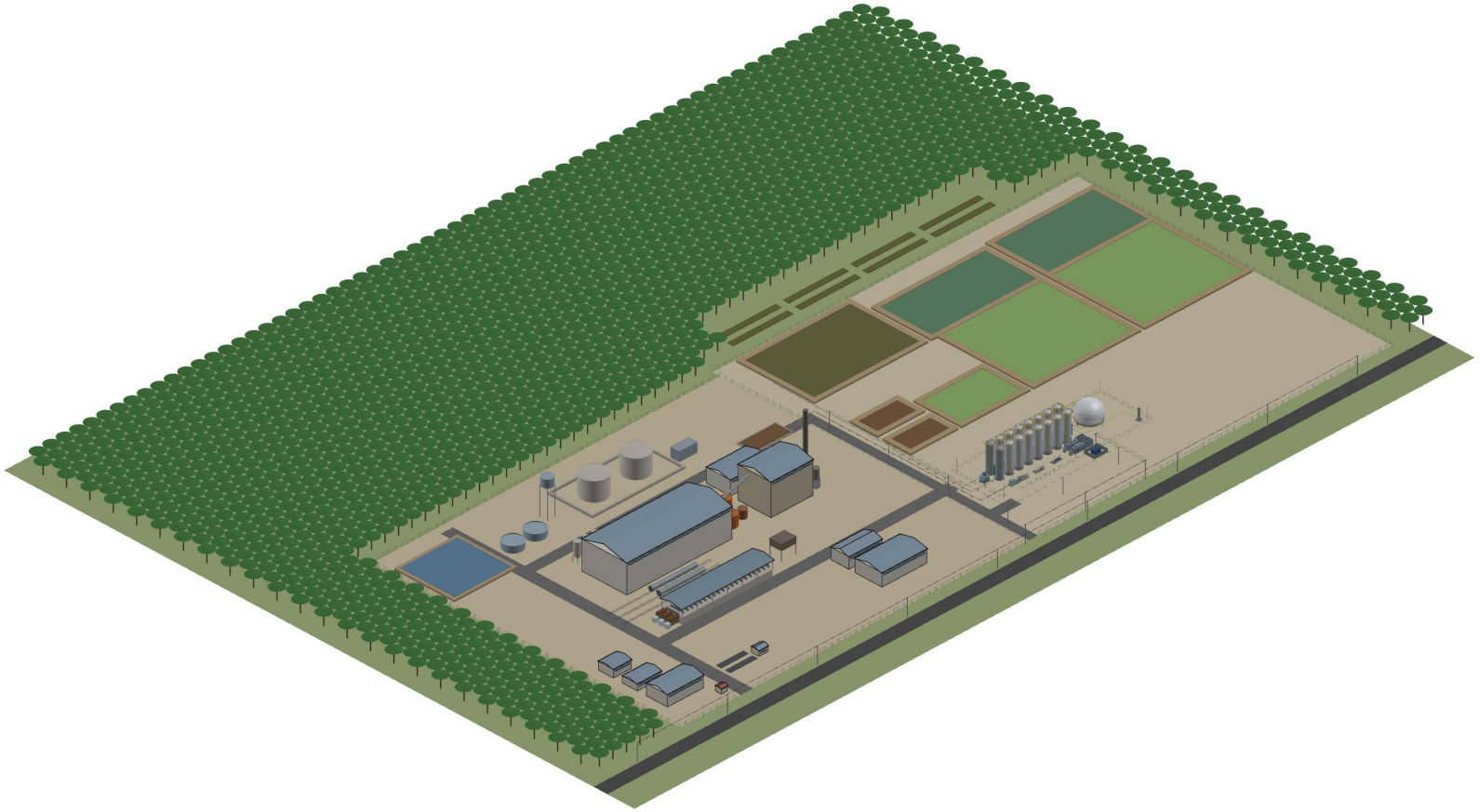
1. Fat pit PKS: POME panas 85 °C, 585 m³/hari
2. PM-1 praolah: saringan, desander, pendingin 85 -> 56 °C
3. 14 reaktor AnMBR termofilik 55 °C, HRT 4,6 hari
4. PM-4 membran UF 996 m²: biomassa kembali ke reaktor
5. GH-301 penampung biogas 1.135 m³ (11.815 Nm³/hari, CH4 58,5%)
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14. Kolam anaerobik lama dibebaskan 1,1 ha; emisi neto 32.288 t CO2e/th dihindari

Garis alir skematis (diperbesar), bukan diameter pipa nyata.

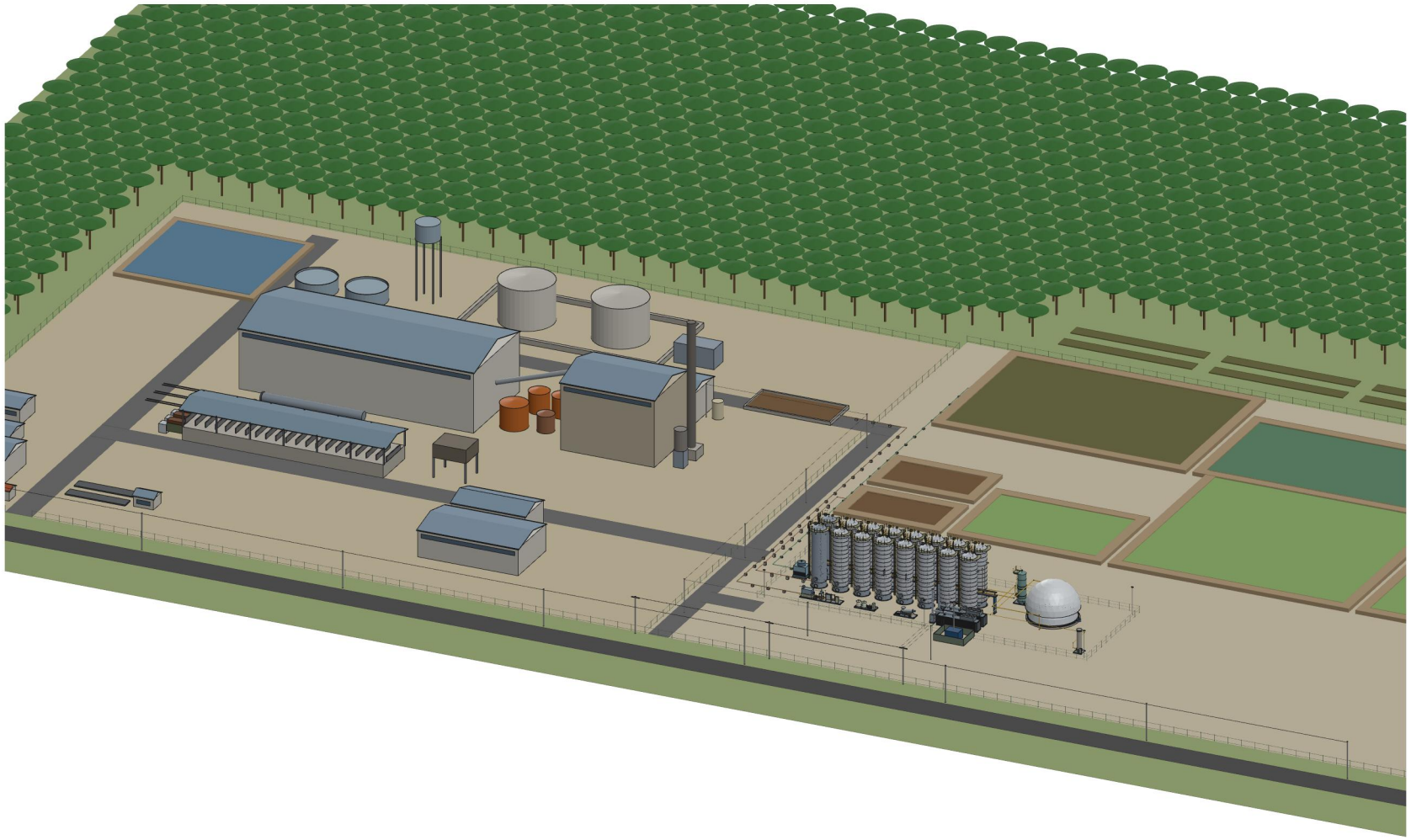
Model 3D lokasi - udara tenggara



Model 3D lokasi - udara barat daya

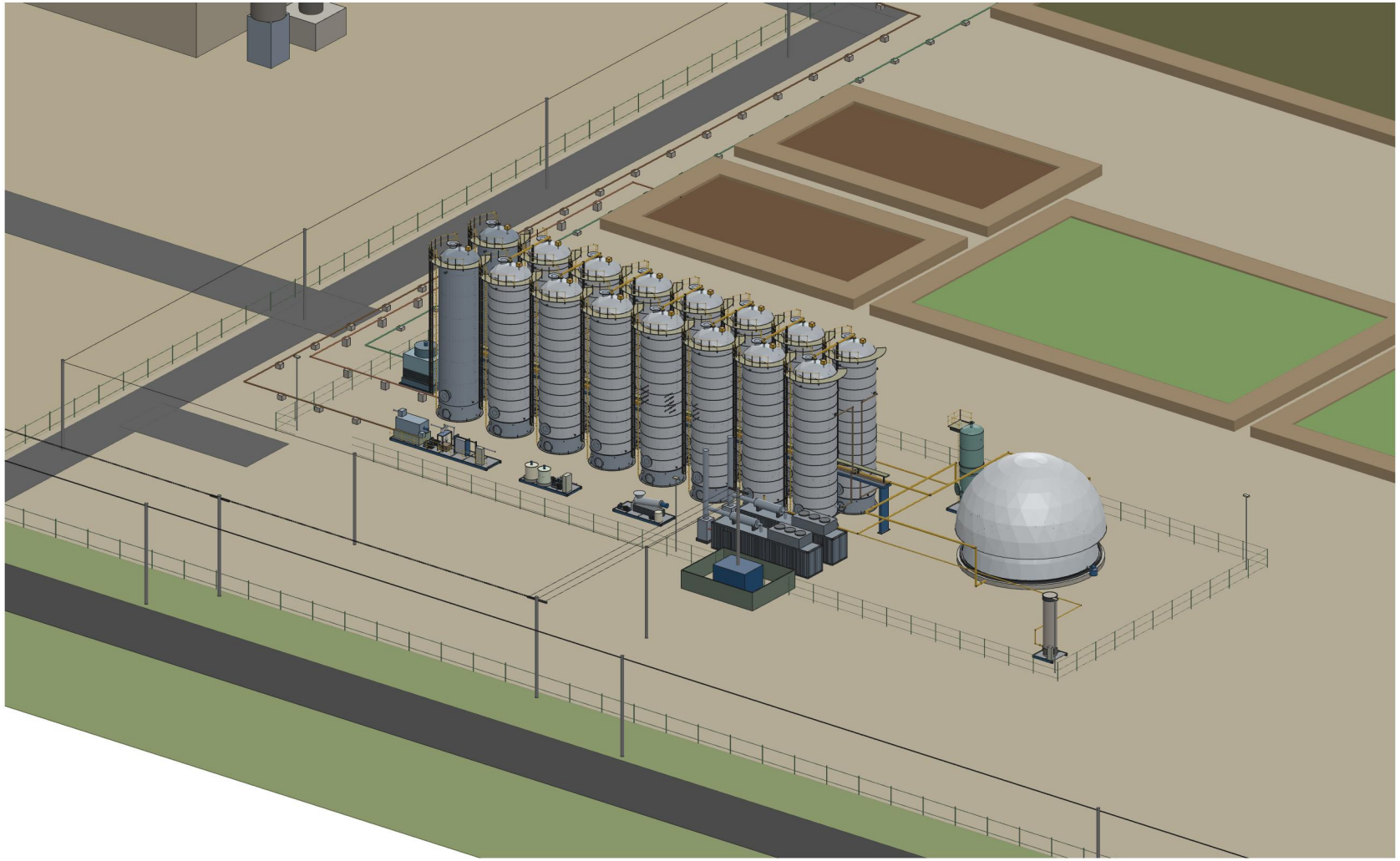


Produk di samping PKS

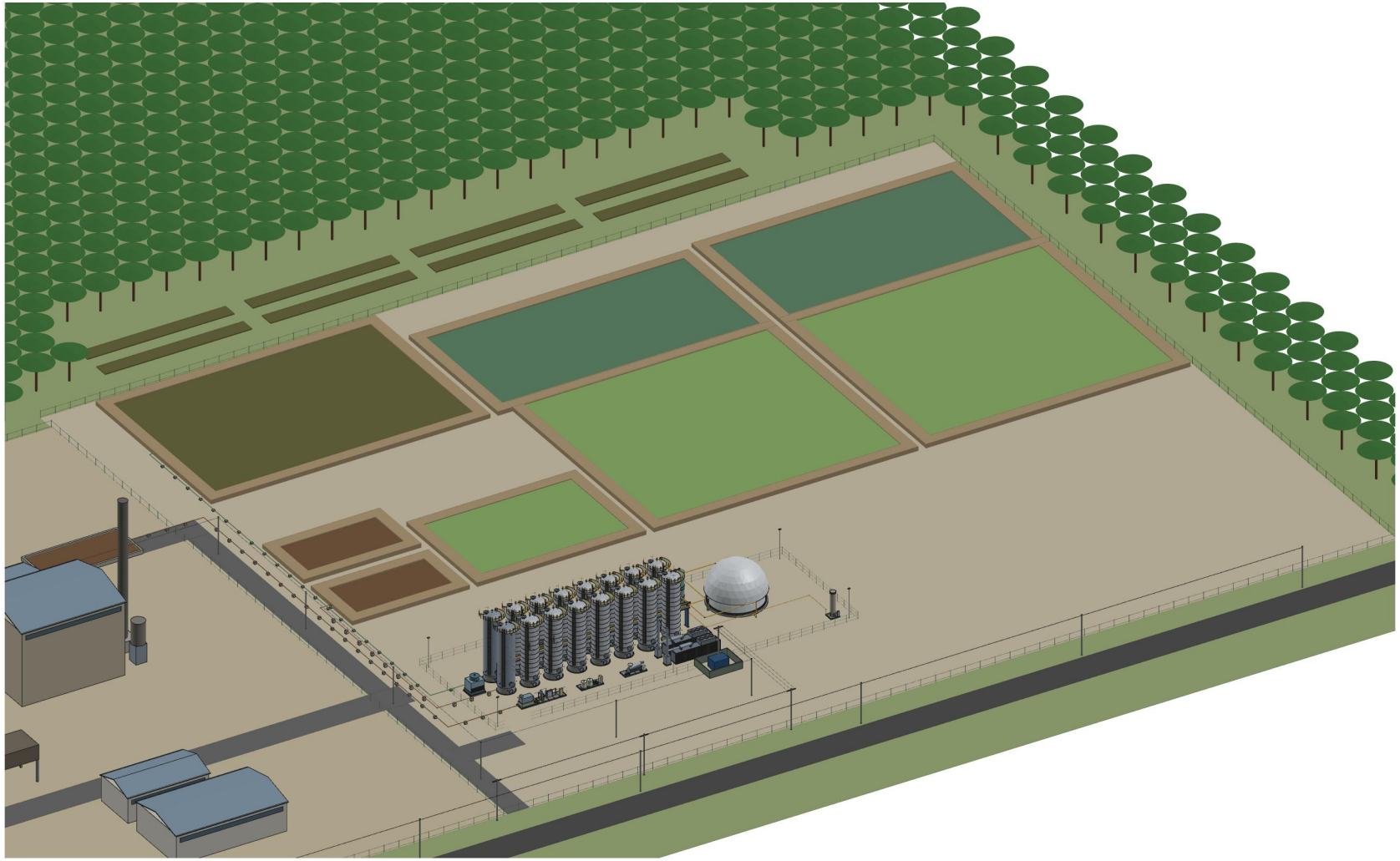


Generator Biogas Portabel - model 3D lokasi (site3d.py), STEP di cad/out/step/SITE_step.zip

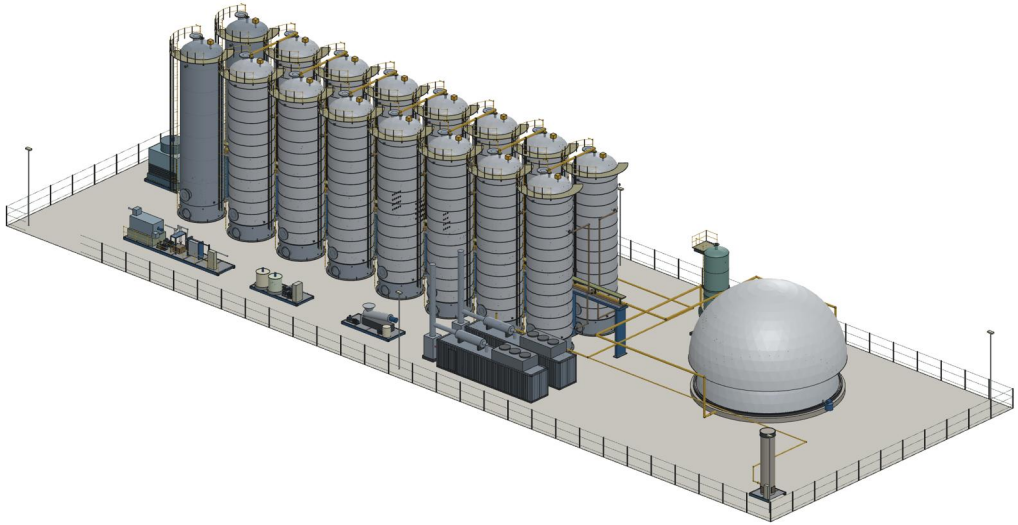
Produk - jarak dekat



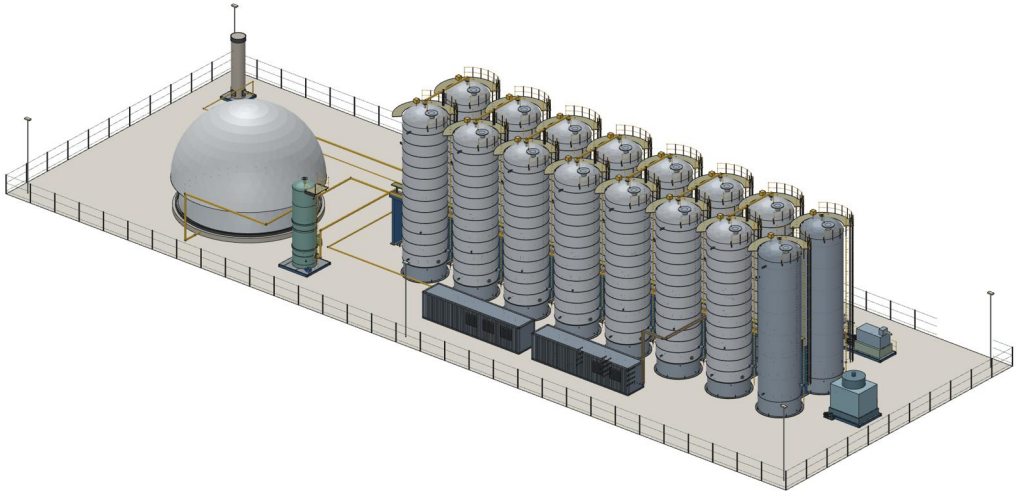
Produk dan sistem kolam



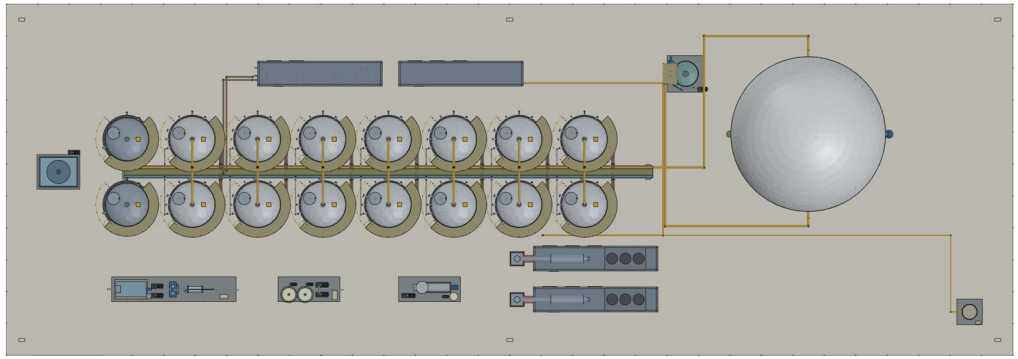
Susunan pabrik 45 t TBS/jam



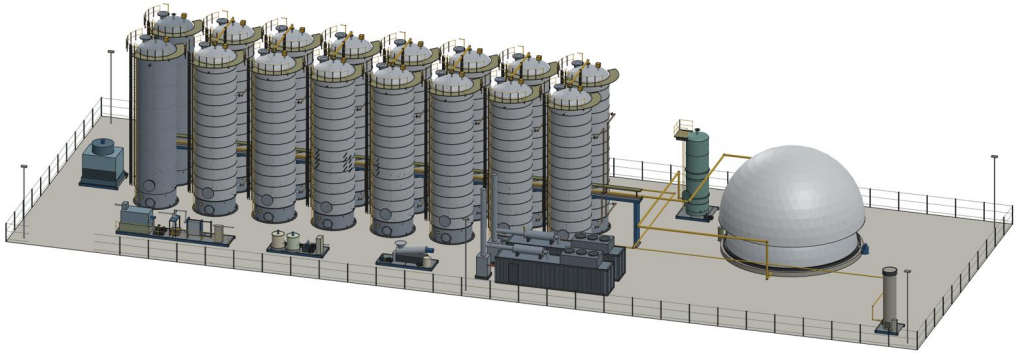
isometri depan



isometri belakang

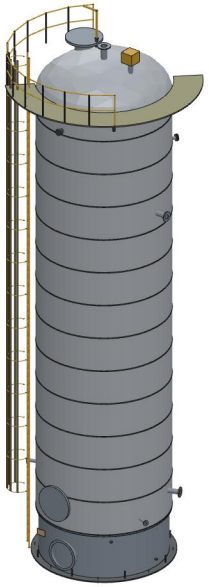


tampak atas

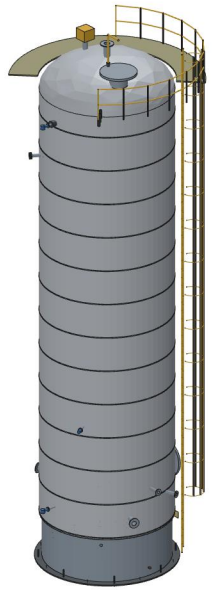


isometri selatan

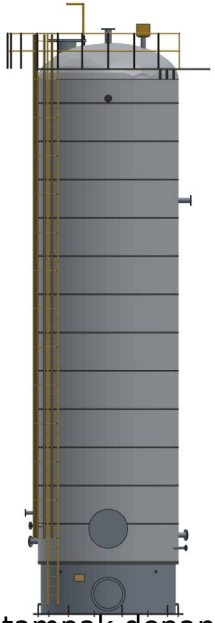
Reaktor AnMBR standar R-201..R-214



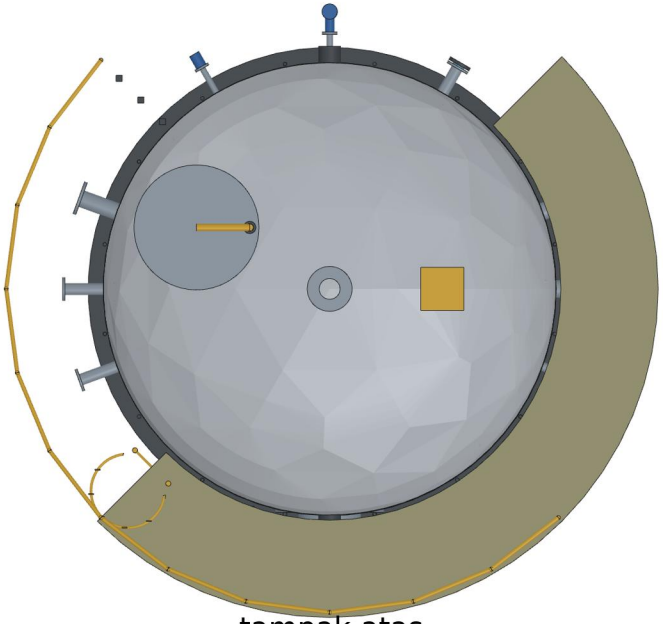
isometri depan



isometri belakang

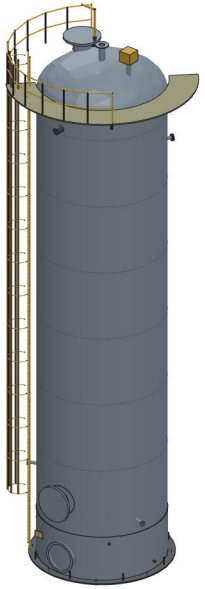


tampak depan

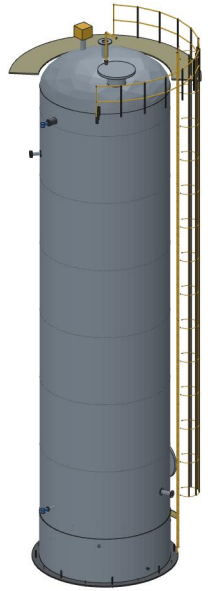


tampak atas

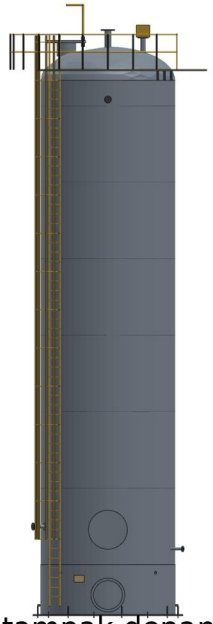
Bejana standar TK-101 / TK-301



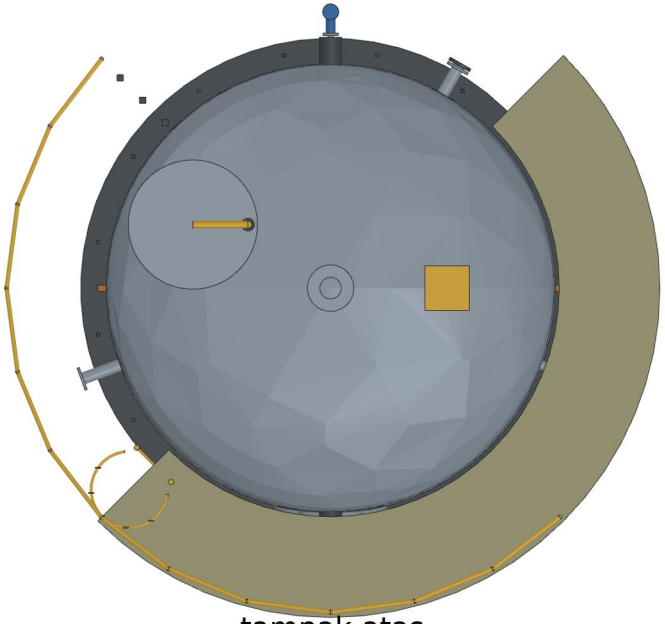
isometri depan



isometri belakang

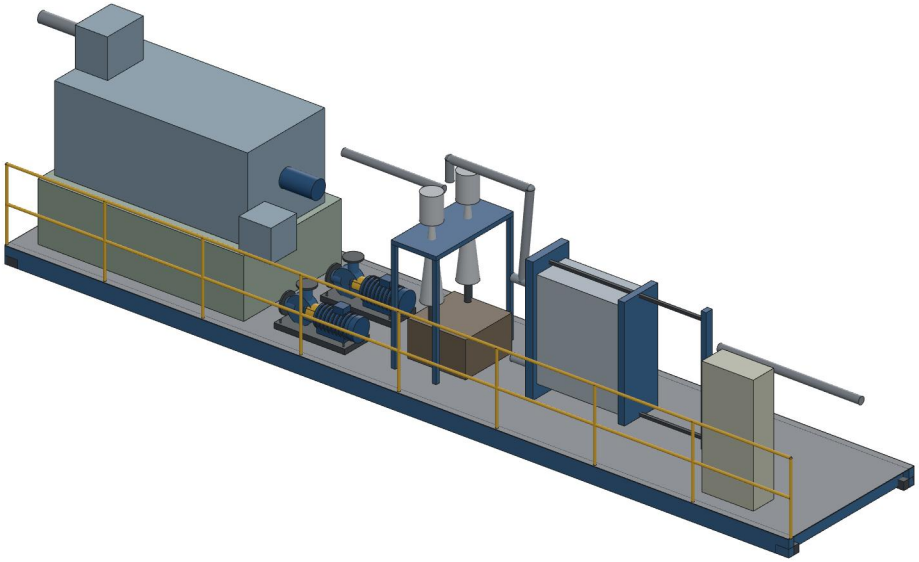


tampak depan

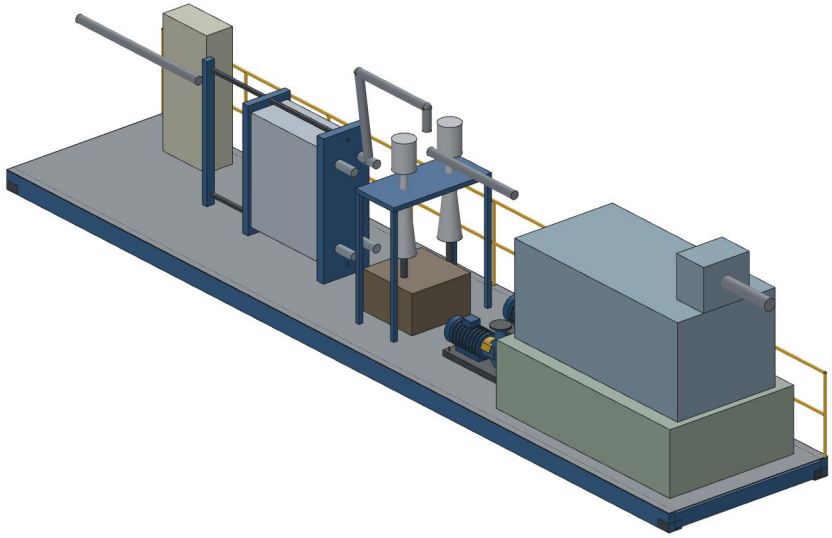


tampak atas

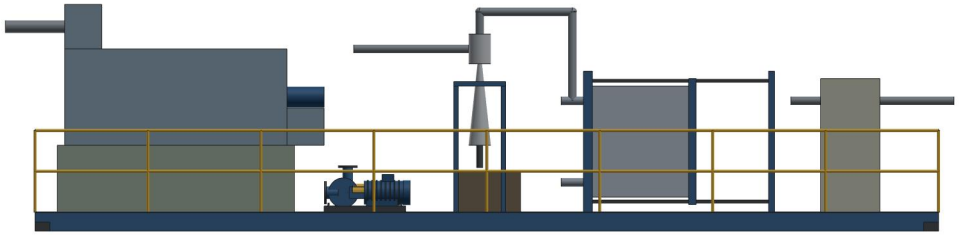
PM-1 skid praolah



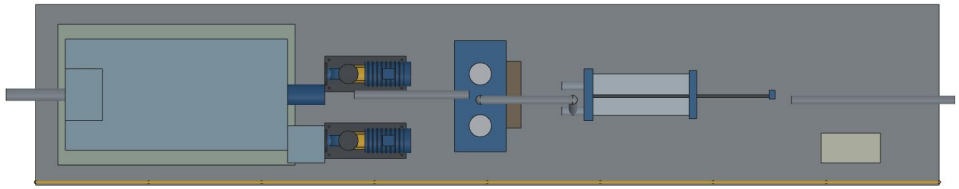
isometri depan



isometri belakang

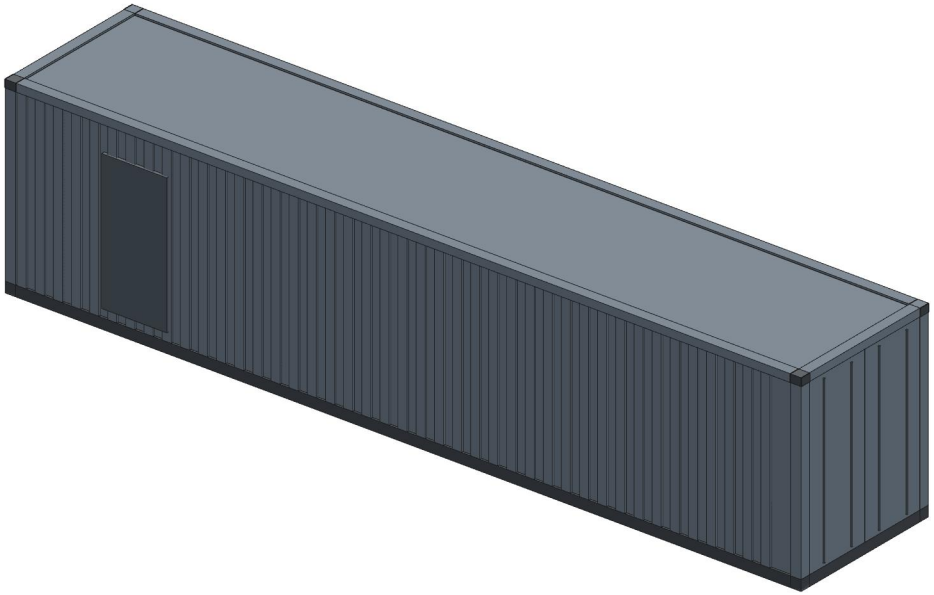


tampak depan

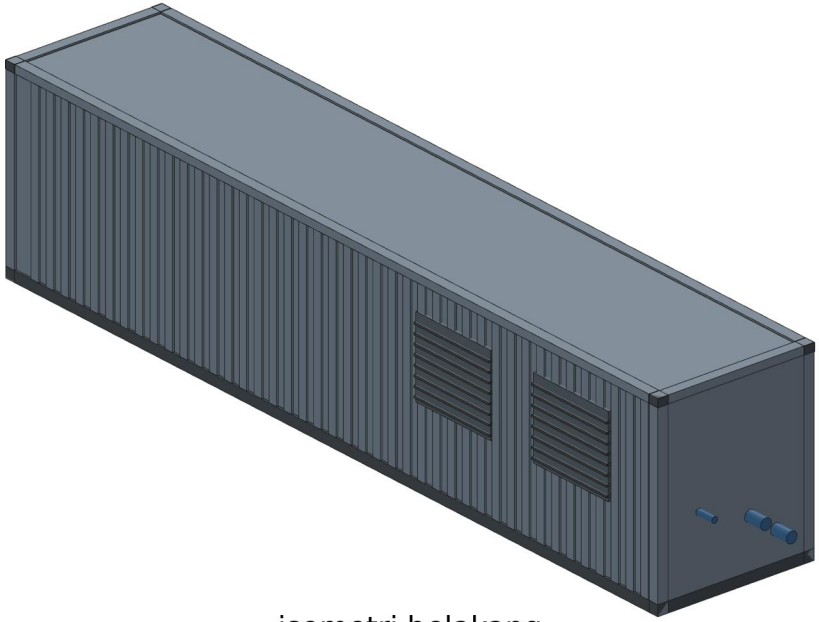


tampak atas

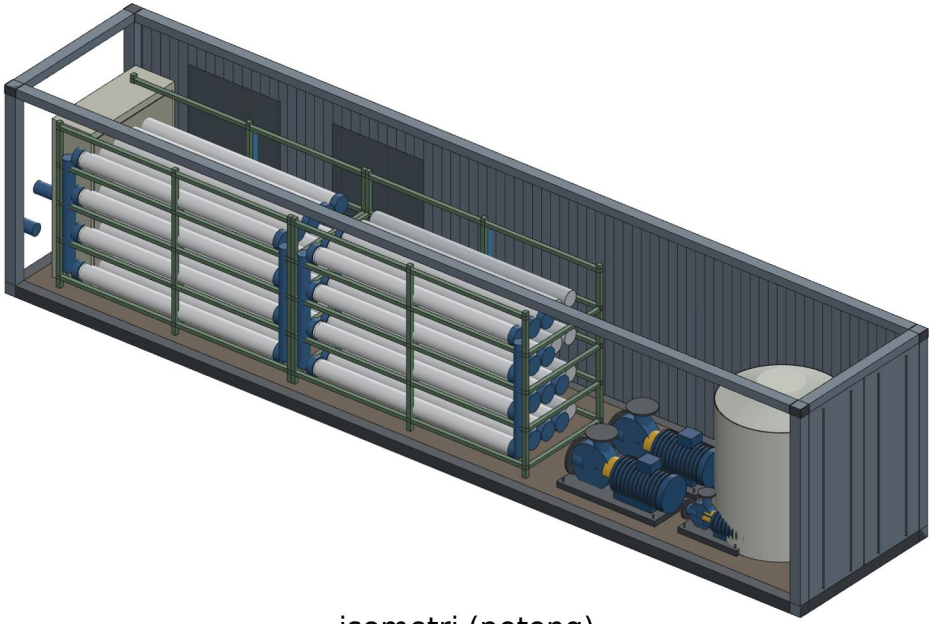
PM-4 kontainer membran



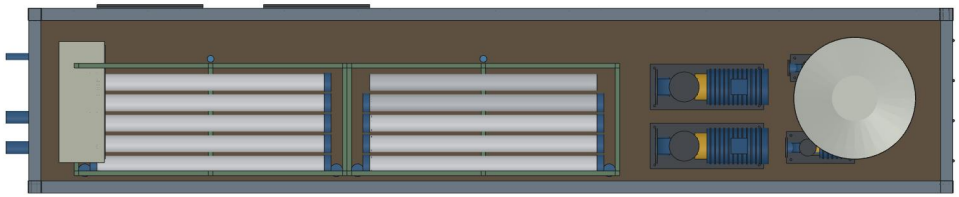
isometri depan



isometri belakang

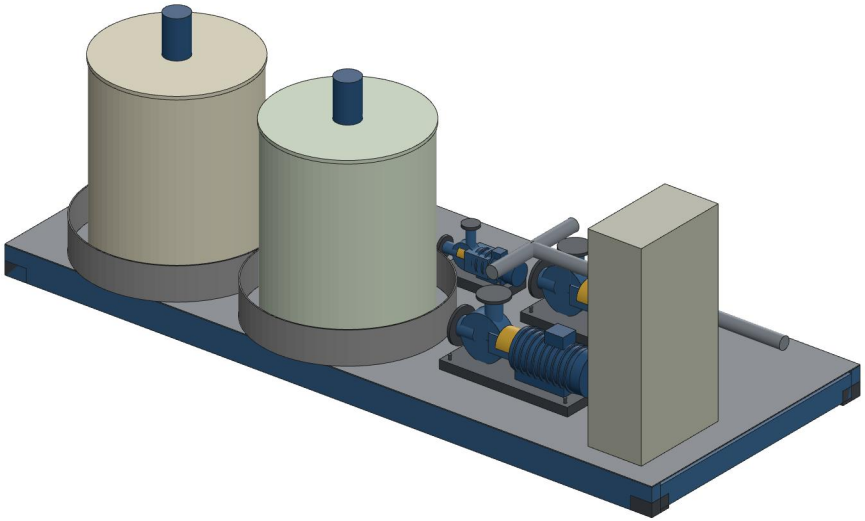


isometri (potong)

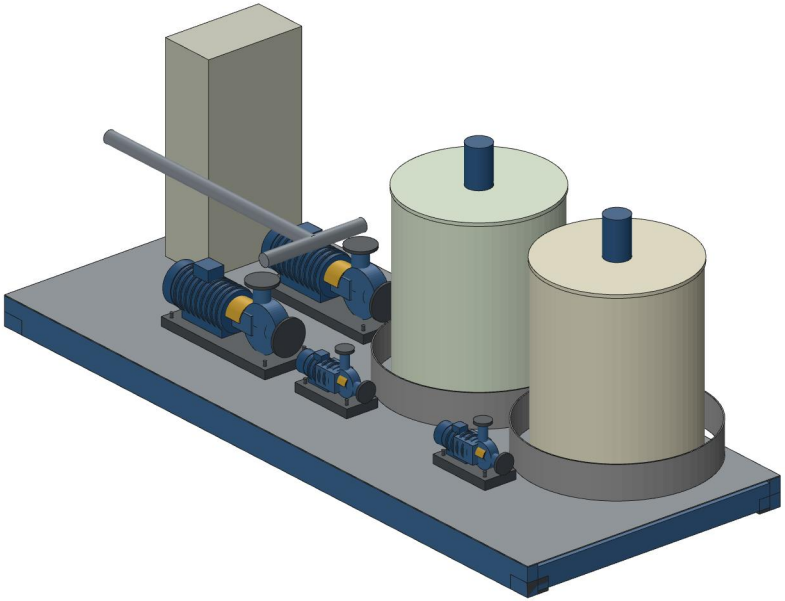


atas (potong)

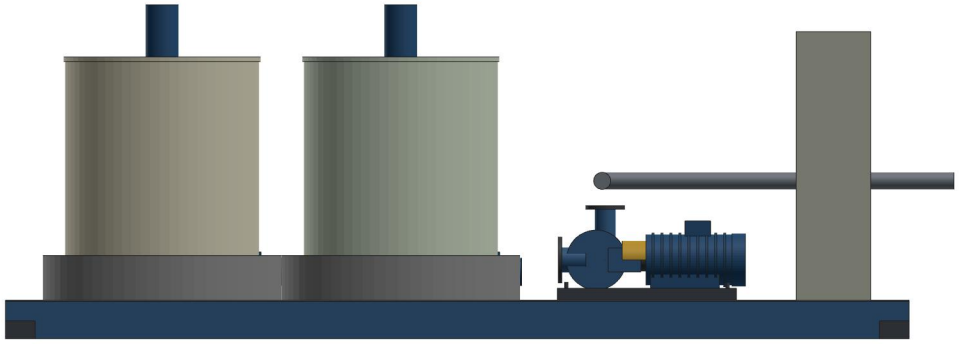
PM-5 skid dosing + pompa umpan



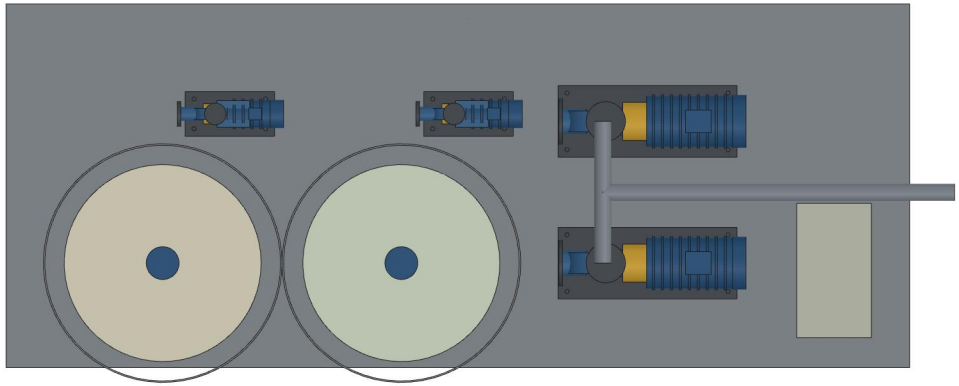
isometri depan



isometri belakang

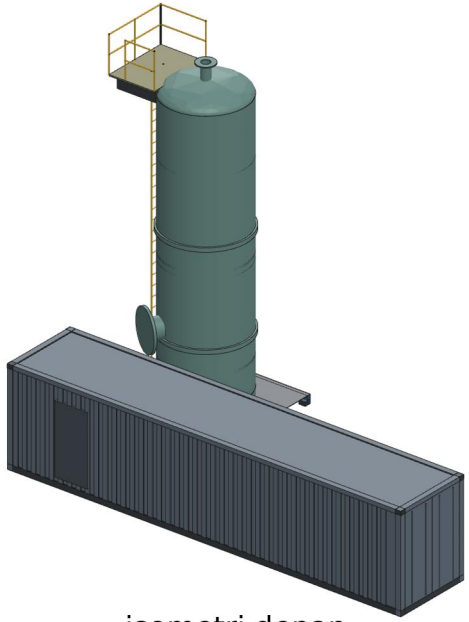


tampak depan

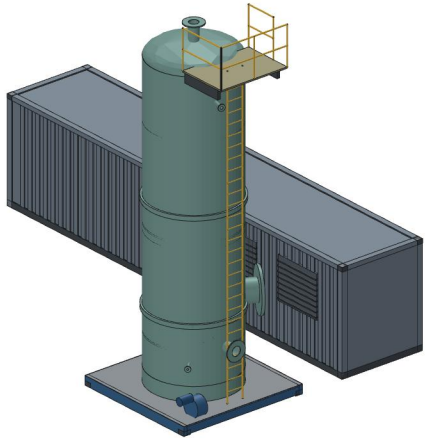


tampak atas

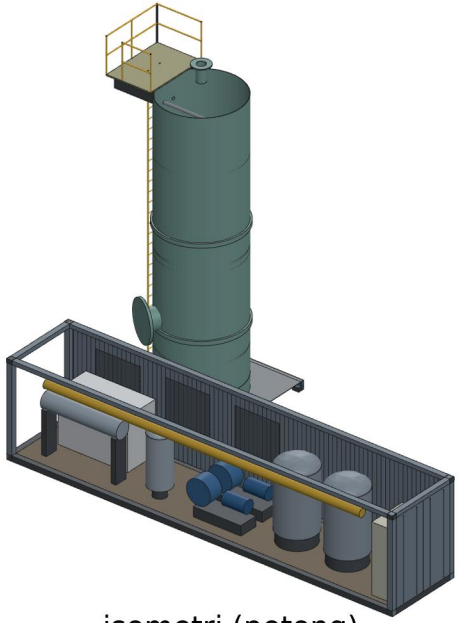
PM-6 skid pengolahan gas



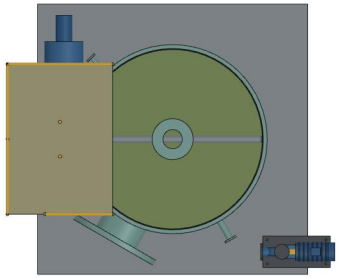
isometri depan

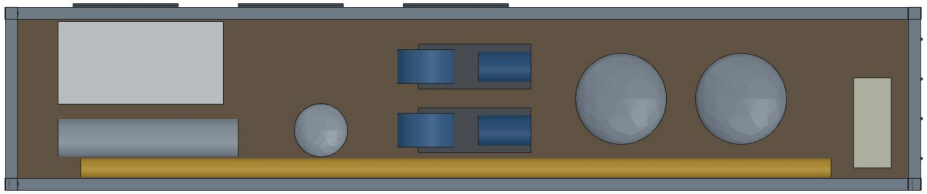


isometri belakang



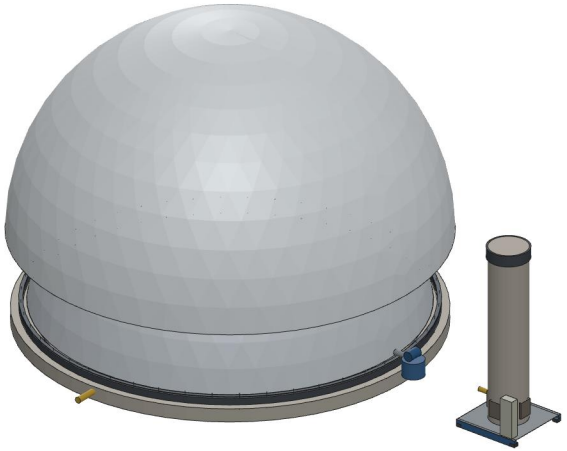
isometri (potong)



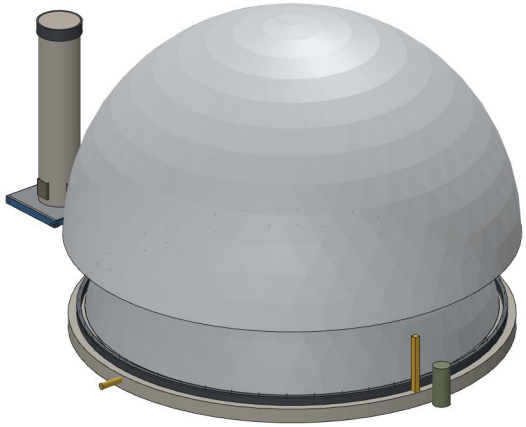


atas (potong)

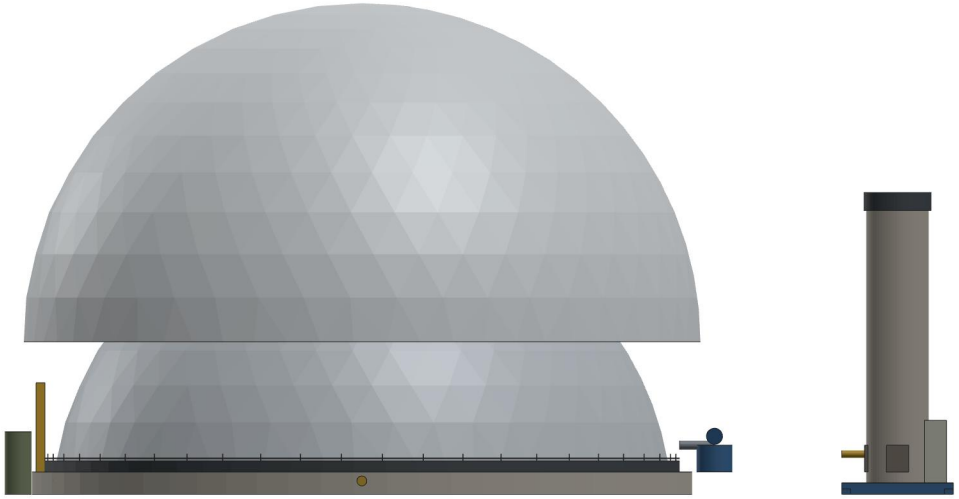
PM-7 penampung gas + flare



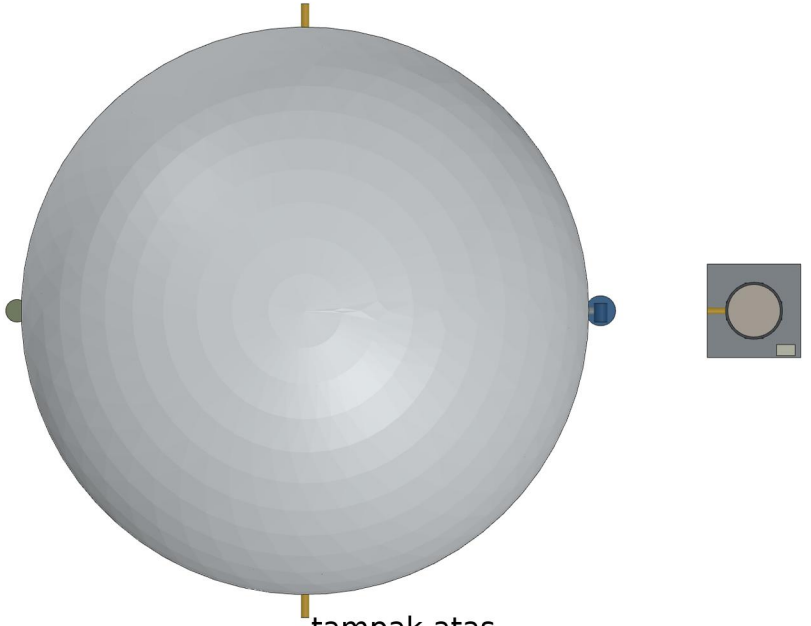
isometri depan



isometri belakang

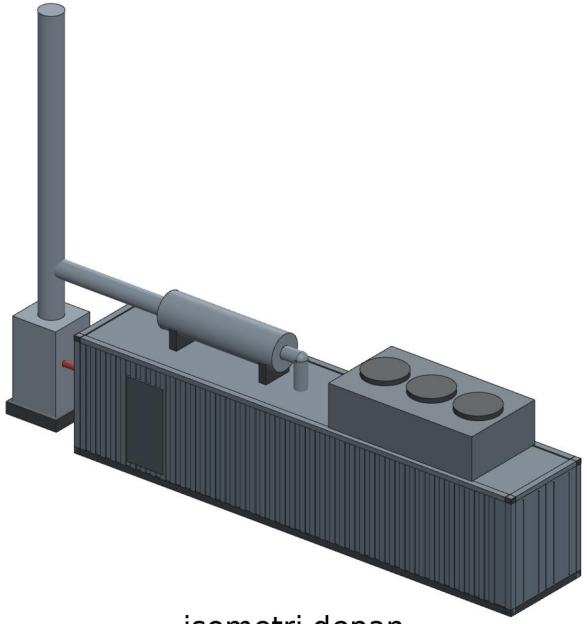


tampak depan

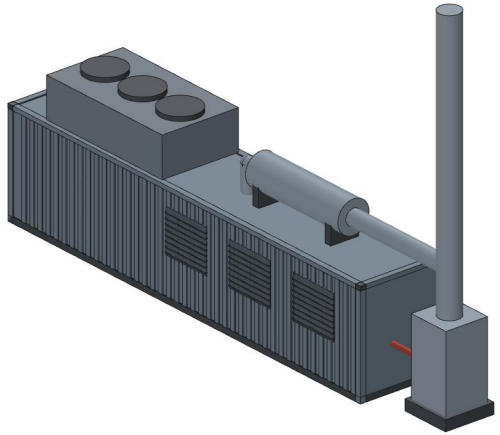


tampak atas

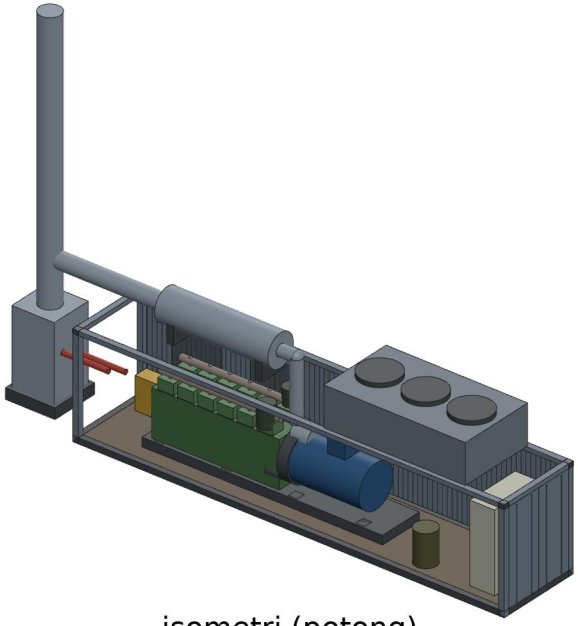
PM-8 kontainer CHP + PM-9 HX-401



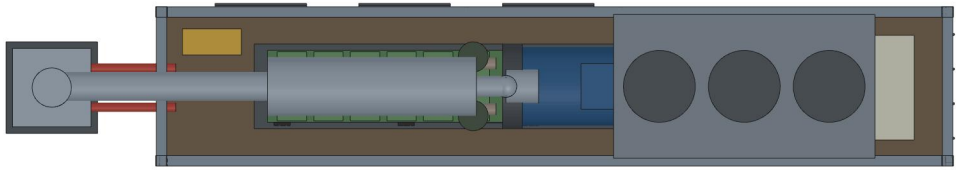
isometri depan



isometri belakang

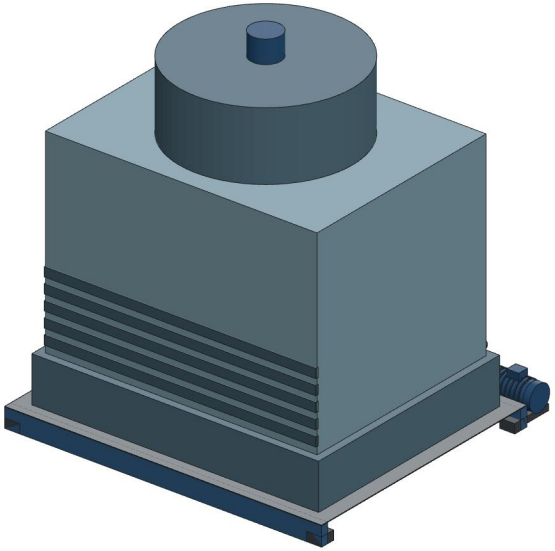


isometri (potong)

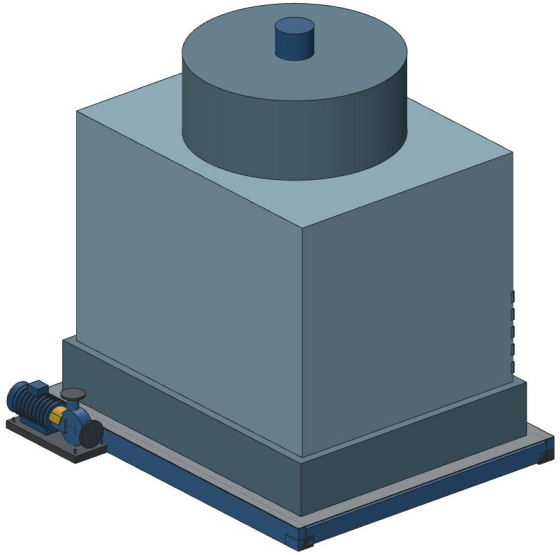


atas (potong)

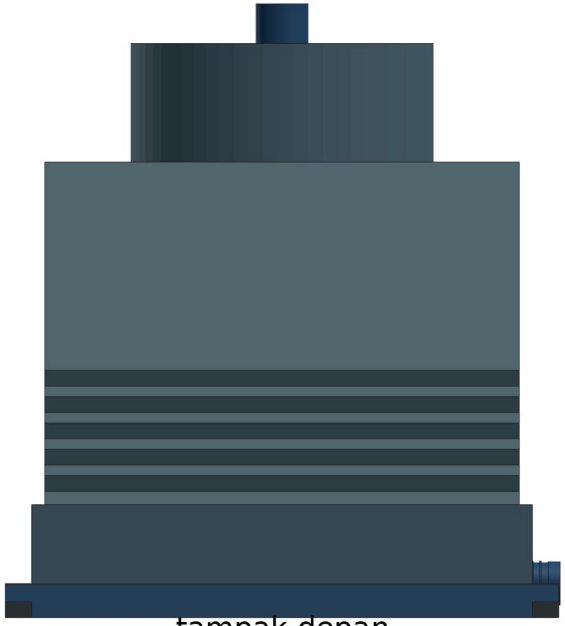
PM-9 menara pendingin



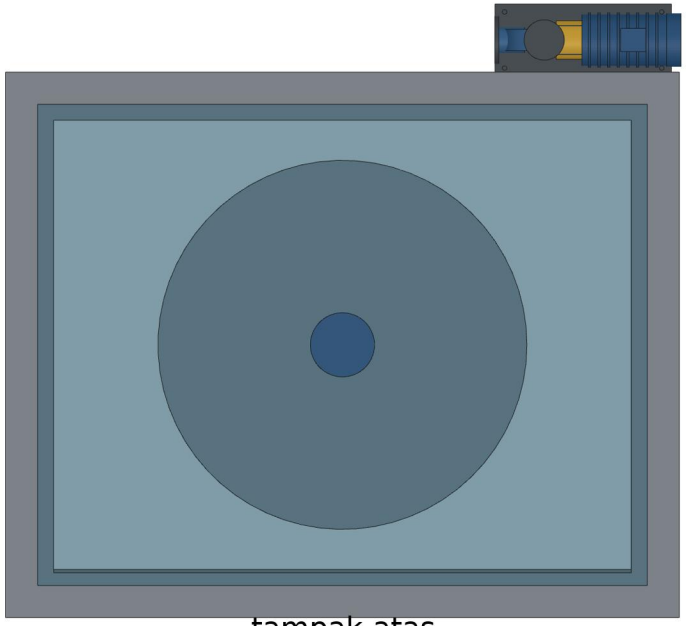
isometri depan



isometri belakang

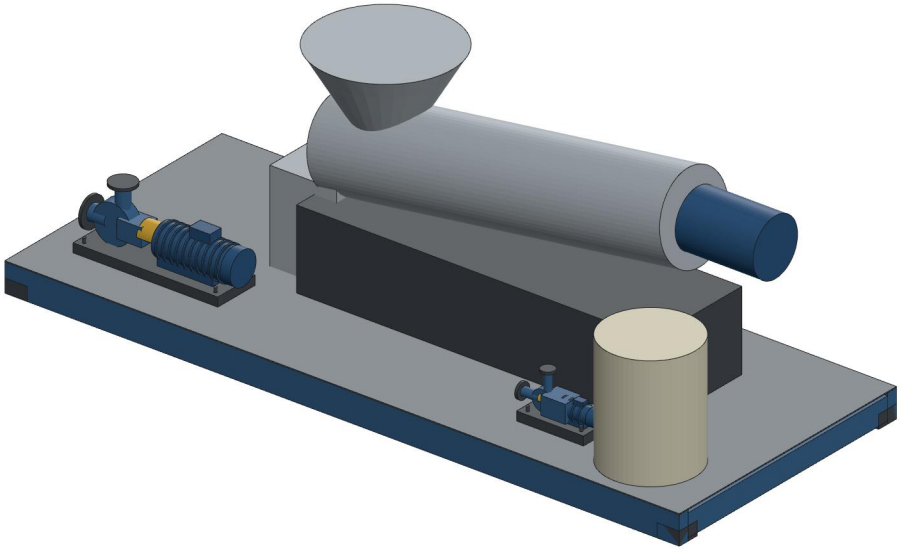


tampak depan

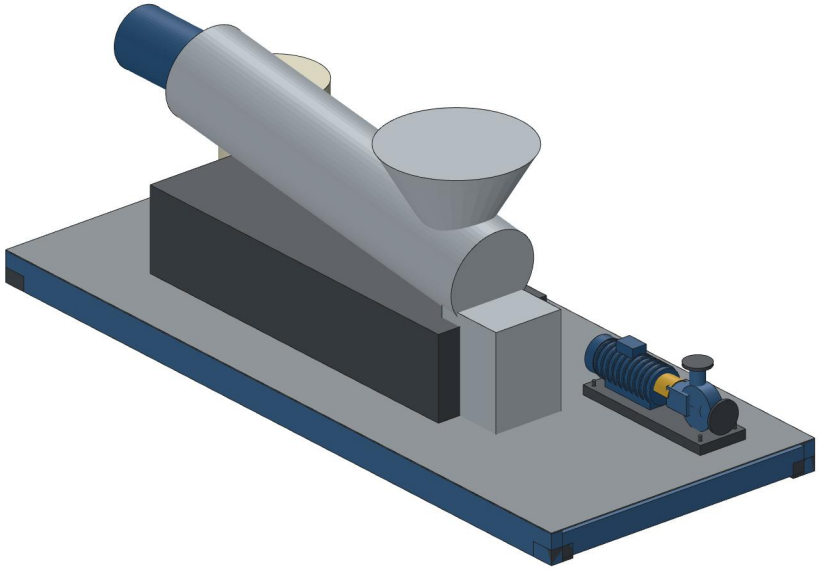


tampak atas

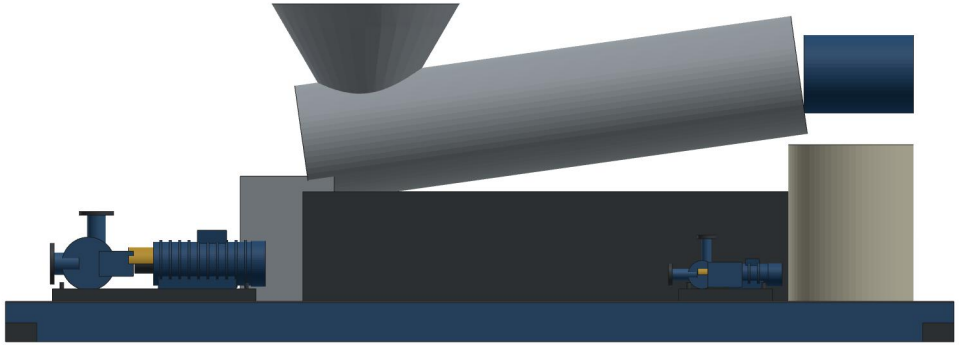
PM-10 skid lumpur



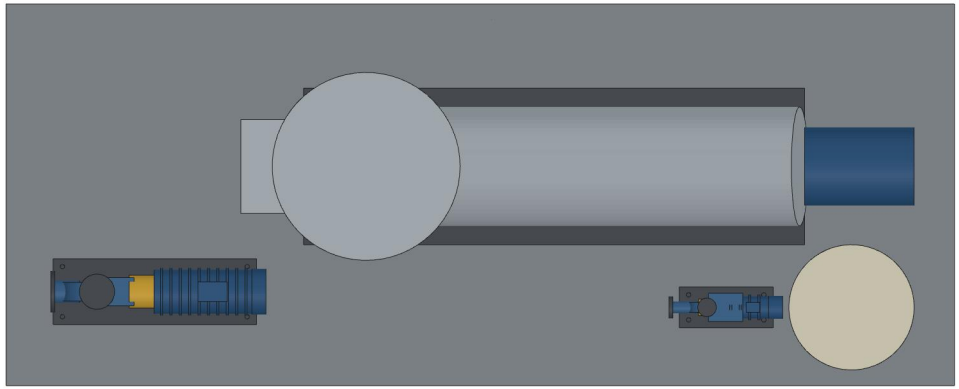
isometri depan



isometri belakang



tampak depan



tampak atas